SOIL EROSION AND CONSERVATION · NOTE ER11

Wind Erosion Control — Crops, Strip Cropping, Mulching, Tillage, Shelterbelts and Sand-Dune Fixing

Course topics: SWCE8.15 L7, SWCE-510 T9, SWCE-505 T3

Principles

1. Cover the soil with vegetation or residues.
2. Build stable aggregates or clods big enough to resist the wind.
3. Roughen the surface to slow the wind and trap drifting soil.
4. Put barriers or trap strips at intervals to cut wind speed and drifting.

Effectiveness varies with climate, soil and land use; cultivated land usually needs a combination.

- **First step — land capability classification.** In western Rajasthan, highly wind-erodible land is unfit for cropping without conservation measures. Much land better suited to permanent grass or forest is still cultivated, giving low returns and severe erosion; keep such land permanently under grass or forest.

Vegetation

Vegetation — intertilled crops, close-growing crops, shrubs and trees, and crop residues — is the most effective control. Cover absorbs part of the wind's force, slows the wind at the surface and transfers the force to anchored plants instead of loose grains.

- **Intertilled crops** give partial protection, depending on row direction relative to the wind, row spacing, plant density, crop type and maturity, soil and climate. Sow rows **perpendicular to the prevailing wind** — with south-west summer winds, run rows SE–NW.
- **Close-growing crops** protect better, depending on growth stage and cover density. Keep land under a dense summer crop whenever possible.

Strip cropping

- Alternate strips of erosion-resistant crops (small grains, closely seeded crops) and susceptible crops (intertilled — potato, cotton, maize, tobacco), with the strips' length across the prevailing erosive wind.
- Strips slow the wind and trap saltating grains, preventing soil avalanching; narrower strips work better.

Soil	Maximum strip width
Sandy	6 m
Loamy sand	8 m
Sandy loam	30 m

Stubble mulching

- Keeping crop residues on the surface protects well against blowing — both during and between crops. Protection depends on the quantity and quality of residue, surface roughness and cloddiness, field size and wind strength.
- Standing residue protects better than leaning residue, and leaning better than flattened; tall better than short. Especially important in drylands and fallow periods.

Soil	Residue needed
Slightly erodible	≈ 600 kg/ha
Moderately susceptible	≈ 2.5 t/ha
Highly erodible dune sand	≥ 10 t/ha

Tillage

- Tillage can increase or decrease blowing. Good tillage leaves a rough, cloddy surface with anchored residues.
- **Deep ploughing** to 30–45 cm improves the wind resistance of sandy soils by burying the erodible surface and bringing up silt and clay — but it must support a permanent vegetation system, or the land degrades in the long run.
- Soil structure depends on the number, depth and speed of operations and the tool angle. Repeated tillage loses residue fast; deep tillage makes more stable clods than shallow. In semi-arid regions do primary tillage soon after rain, with the fewest operations to avoid pulverising the soil.

Implement	Effect
Sweep-type furrow openers	Suitable — leave residues erect
One-way disc plough (properly angled)	Suitable — leaves part of the residue standing
Lister	Suitable — creates the greatest surface roughness
Subsurface tillage with straight or V-blades	Suitable — retains 80–90% of residue
Mouldboard plough	Unsuitable — buries residues
Disc harrow and vertical-disc tools	Unsuitable — break clods and bury residues

Surface roughness and moisture conservation

- Soil movement falls as roughness rises. Ridges — even plough furrows — at right angles to the wind reduce soil loss; large clods and irregularities add roughness. Contour bunds also reduce wind erosion, conserve moisture and control water erosion.
- **Moisture conservation** is crucial in arid and semi-arid regions: increase infiltration, reduce evaporation and prevent unwanted plant growth — by contour farming, contour bunding, mulching and suitable crops. More mulch conserves more moisture.

Windbreaks and shelterbelts

- A **windbreak** is any vegetative barrier that reduces wind speed. A **shelterbelt** is rows of trees and shrubs at right angles to the wind that protect soil, moisture and crops.

- Effectiveness depends on wind speed and direction and on the barrier's shape, width, height and porosity.

Position (H = belt height)	Wind-speed reduction
Near the belt and just leeward	60–80%
About 20 H leeward	≈ 20%
Beyond 30–40 H leeward	None
Windward side	Protection extends 5–10 H

- Stronger winds need closer belt spacing. For full protection a windbreak should be at least 10 × its height long.
- **Benefits:** less energy for heating and cooling homes, a better crop microclimate, wood at harvest, alley cropping, less noise and less evaporation.
- **Ideal belt:** a pyramidal profile — shrubs on the windward side, medium trees in the middle, tall trees leeward; 10, 7, 5, 4, 3 or 2 rows; minimum length ≈ 24 × height; width typically 12–20 m; no gaps, with tapered ends.

Type	Recommended species
Grasses	*Saccharum spontaneum*, *S. munja*, *Panicum antidotale*, *Cenchrus* spp.
Shrubs	*Calotropis procera*, *Clerodendron phlomoides*, *Cassia auriculata*, *Dodonaea viscosa*
Trees	*Acacia arabica*, *A. leucophloea*, *Dalbergia sissoo*, *Prosopis juliflora*, *Tamarix articulata*, *Parkinsonia aculeata*, *Casuarina equisetifolia*, etc.

Fixing sand dunes

1. Level the sand crests.
2. Cross-till with a lister, heavy plough or deep cultivator.
3. Seed fast-growing grasses — in a calm, high-rainfall period, starting from the windward side.
4. On large dunes, reseed progressively each year until the cover is complete, then plant permanent grasses, shrubs or trees.
5. Stabilise further with straw or brush barriers, and protect from fire, overgrazing and tree cutting.

Western Rajasthan ecosystem	Grasses
Rugged hills	*Sehima nervosum*, *Eremopogon foveolatus*, *Heteropogon contortus*
Hill folds and stabilised dunes	*Dichanthium annulatum*
Sand plains and shifting dunes	*Elyonurus hirsutus*, *Cenchrus ciliaris*, *C. setigerus*, *C. prieurii*
Saline/alkaline soils	*Cenchrus ciliaris*, *C. setigerus*, *C. prieurii*, *Sporobolus arabicus*

Trees for dune stabilisation: *Tecoma undulata* (rohira; now *Tecomella undulata*), *Prosopis cineraria* (khejri), *Prosopis juliflora*, *Azadirachta indica* (neem), and some eucalyptus and acacia species suited to arid India. (The notes spell the first hill grass "Schima"; the genus is *Sehima*.)

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. ICAR–Central Arid Zone Research Institute (CAZRI), Jodhpur — sand-dune stabilisation and shelterbelt technology. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Rows of intertilled crops should be sown

(a) parallel to the prevailing wind
(b) perpendicular to the prevailing wind
(c) diagonally
(d) in any direction

Q2. Maximum strip width for wind-strip cropping on sandy soil is

(a) 6 m
(b) 8 m
(c) 30 m
(d) 60 m

Q3. Residue needed to protect highly erodible dune sand is at least

(a) 600 kg/ha
(b) 2.5 t/ha
(c) 5 t/ha
(d) 10 t/ha

Q4. Which implement leaves the greatest surface roughness?

(a) mouldboard plough
(b) disc harrow
(c) lister
(d) roller

Q5. Subsurface tillage with V-blades retains about ___ of residue

(a) 20–30%
(b) 50%
(c) 80–90%
(d) 100%

Q6. A shelterbelt gives no wind reduction beyond about ___ belt heights leeward

(a) 5–10
(b) 10–15
(c) 20
(d) 30–40

Q7. Windward protection of a shelterbelt extends about

(a) 1–2 H
(b) 5–10 H
(c) 20 H
(d) 40 H

Q8. Dune stabilisation work should start from the

(a) leeward side
(b) windward side
(c) crest
(d) middle

Q9. Typical shelterbelt width is

(a) 2–5 m
(b) 12–20 m
(c) 50–100 m
(d) 200 m

Answers

Q1 — (b) perpendicular to the prevailing wind
Q2 — (a) 6 m
Q3 — (d) 10 t/ha
Q4 — (c) lister
Q5 — (c) 80–90%
Q6 — (d) 30–40
Q7 — (b) 5–10 H
Q8 — (b) windward side
Q9 — (b) 12–20 m